Subroutines: user-defined functions and procedures

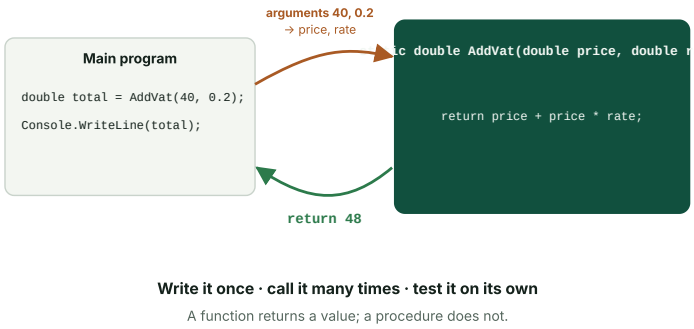
D19 FACT FILE

Specification: students should be able to

- Building reusable code that utilises user-defined functions or methods

CCEA fact file: Programming Constructs 3

Key words: subroutine, procedure, function, method, parameter, argument, return value, local variable, modular, reusable, built-in function



Write it once · call it many times · test it on its own

A function returns a value; a procedure does not.

A main program passing arguments into a function and receiving the return value

Subroutines

A subroutine is a self-contained, named block of code for one task, defined once and called whenever needed. A procedure performs a task; a function also returns a value to the calling code. Data is passed in through parameters. Variables created inside a subroutine are local to it.

Benefits

Subroutines avoid repeating code, make programs shorter and easier to read, and allow each part to be tested and debugged independently. Changes are made in one place. They support decomposition and team working, and well-written subroutines can be reused in other programs.

T Key terms from the mark schemes

Function
A named subroutine that performs a task and returns a value.
Credited: U4 2022 Q7, 2023 Q4

Procedure
A named subroutine that performs a task without returning a value.
Credited: U4 2022 Q7

Parameter
A value passed into a subroutine.

C C# example

Parameters and return values

```
static double AddVat(double price, double rate)
{
    return price + price * rate;
}

double total = AddVat(40, 0.2);
Console.WriteLine(total);      // 48
```

Exam focus

Define a function and give benefits (2022 Q7(b)), two advantages of user-defined functions (2023 Q4(c)(iii)). Unit 5 solutions are expected to use user-defined functions.

Unit 4 2023 Q4 [23]	Decomposition and abstraction; flowchart symbols; syntax/logic errors; functions; menu validation loop
Unit 4 2022 Q7 [18]	Array indexing; weekly totals and average algorithm; functions and their benefits

? Check yourself

1. Define the term function. [2]
2. Describe the difference between a function and a procedure. [2]
3. What is a parameter? [1]

Answers: 1. A named, self-contained block of code that carries out a specific task (1) / and returns a single value to the code that called it (1) 2. A function returns a value to the calling code (1) / a procedure carries out a task but does not return a value (void in C#) (1) 3. A value/variable passed into a subroutine that it needs to carry out its task, listed in its definition (1)